

Qidong Sun

Undergraduate in Robotics Engineering, Chongqing University (Project 985 University)

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Education

Chongqing University — Elite Institute of Engineering

2023 – Present

B.Eng., Robotics Engineering (Mingyue Innovation Class)

GPA 3.49/4.00; Weighted Average 85.81/100 .

Selected coursework: Mathematics and Physics III (96/100); Foundations of Robotics (94/100); Principles of Engineering (93/100); Quantitative Engineering Design II (92/100).

Visiting Scholarship

China Scholarship Council (CSC) fully funded visiting scholarship, **Singapore University of Technology and Design (SUTD)**.

Expected visiting period: September – December 2026.

Selected Research

VLA-Based Dual-Arm Manipulation Pipeline via pi-0 in NVIDIA Isaac Sim

CQU Embodied Intelligence Lab 09/2025 – 03/2026

- Developed a dual-arm simulation pipeline in NVIDIA Isaac Sim to collect multimodal demonstrations, including RGB observations, language instructions, and proprioceptive joint states.
- Implemented a WebSocket-based client to connect a pi-0 VLA model with the simulator for closed-loop observation–action communication.
- Engineered an automated data ingestion workflow converting raw simulation episodes into LeRobot-style intermediate formats and serialized Parquet datasets.
- Built dataset validation tools to check action–observation consistency, joint-name alignment, and trajectory logging.

SO-101 Manipulator Integration for Demonstration Data

CQU Embodied Intelligence Lab 03/2026 – present

- Building a MuJoCo-based simulation workflow for the SO101 robotic arm to study learning-based manipulation control.
- Running pick-and-lift experiments with Gymnasium-style environments, focusing on observations, action spaces, reward signals, and success conditions.
- Preparing an imitation learning baseline and planning RL extensions to compare policy learning performance under different demonstrations, rewards, and simulation settings.

Selected Projects

RoboMaster Decision Stack Refactor (ROS 2, C++)

GitHub 07/2024 – 05/2025

- Refactored the autonomy stack into state estimation, decision-making, and action execution layers, improving modularity and scalability.
- Built standardized ROS 2 messaging interfaces for perception, decision, and navigation modules.
- Designed a flexible architecture supporting FSM, behavior trees, and future learning-based policies (RL/imitation learning).
- Tested navigation modules in Gazebo and ensured stable communication for real-robot deployment.

ROS 2 Vacuum Cleaning Robot: Navigation Stack and Modular Sensors

GitHub 09/2025 – 12/2025

- Delivered simulation bring-up with robot description assets, Gazebo worlds, navigation and SLAM parameter sets, RViz layouts, and launch scripts that start the full stack together.
- Implemented a swappable two-dimensional lidar front end so simulated scans and live hardware feeds drive the same navigation configuration after a runtime mode switch.
- Ran simultaneous mapping and navigation in Gazebo and on the physical vacuum platform, exported occupancy maps from test runs, and iterated navigation and mapping gains from hardware experiments.

Technical Skills

Programming: Python, C++, MATLAB. **Robotics and simulation:** ROS 2, NVIDIA Isaac Sim, Gazebo, MuJoCo, LeRobot, RViz2, Nav2, SLAM Toolbox. **Tools:** Linux/Ubuntu, Git, CMake, WebSocket.

Honors & Awards

- CSC fully funded visiting scholarship, SUTD, 2026
- National Third Prize, RoboMaster Robotics Competition (National final)
- Regional Second Prize, RoboMaster Robotics Competition, Central China Regional
- Regional Third Prize, RoboMaster University League, Southwest Station
- Successful Participant, Mathematical Contest in Modeling (MCM), 2026